

1 Load necessary packages

```
library(readr)
library(stats)
library(MVN)
library(REdaS)

## Loading required package: grid

library(psych)
library(kableExtra)
library(dplyr)

##
## Attaching package: 'dplyr'
## The following object is masked from 'package:kableExtra':
##
##   group_rows
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(lavaan)

## This is lavaan 0.6-15
## lavaan is FREE software! Please report any bugs.
##
## Attaching package: 'lavaan'
## The following object is masked from 'package:psych':
##
##   cor2cov

library(ggplot2)

##
## Attaching package: 'ggplot2'
## The following objects are masked from 'package:psych':
##
##   %+%, alpha
```

2 data importation

```
Depression <- read_csv("C:/Users/Administrator/Desktop/Depression.csv")

## Rows: 247 Columns: 12
## -- Column specification -----
## Delimiter: ", "
## dbl (12): Index, Gender, Age, x1, x2, x3, x4, x5, x6, x7, x8, x9
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this
message.

#View(Depression)
names(Depression)

[1] "Index" "Gender" "Age" "x1" "x2" "x3" "x4" "x5" [9] "x6" "x7" "x8" "x9"

df<-as.data.frame(Depression)
```

```

#removing the index column
names(df)

## [1] "Index" "Gender" "Age" "x1" "x2" "x3" "x4" "x5"
## [9] "x6" "x7" "x8" "x9"

Depr<-df %>%
  select(-(Index))

col_rem<-c("Gender", "Index")
Depr<-Depression %>%
  select(-one_of(col_rem))

View(Depr)

#removing the index,gender,age columns

cl_rm<-c("Index", "Gender", "Age")
data_dep<-Depression %>%
  select(-one_of(cl_rm))
View(data_dep)

#identifying missing values
sum(is.na(data_dep))

## [1] 0

```

4 Exploratory data analysis (EDA)

- Data Summarization

```

str(Depression)

## spc_tbl_ [247 x 12] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ Index : num [1:247] 1 2 3 4 5 6 7 8 9 10 ...
## $ Gender: num [1:247] 2 1 2 1 1 1 2 2 2 2 ...
## $ Age : num [1:247] 22 35 22 25 24 24 25 20 20 24 ...
## $ x1 : num [1:247] 1 3 1 1 3 1 3 1 3 1 ...
## $ x2 : num [1:247] 0 1 1 1 2 1 1 3 1 1 ...
## $ x3 : num [1:247] 0 1 1 1 3 1 1 2 0 0 ...
## $ x4 : num [1:247] 1 1 2 1 0 0 2 1 1 3 ...
## $ x5 : num [1:247] 0 2 3 0 3 0 1 0 0 1 ...
## $ x6 : num [1:247] 0 2 2 3 3 1 1 3 1 1 ...
## $ x7 : num [1:247] 0 0 0 1 1 0 1 1 1 0 ...
## $ x8 : num [1:247] 1 2 1 1 1 0 2 1 0 1 ...
## $ x9 : num [1:247] 0 0 1 0 3 0 0 0 0 0 ...
## - attr(*, "spec")=
## .. cols(
## .. Index = col_double(),
## .. Gender = col_double(),
## .. Age = col_double(),
## .. x1 = col_double(),
## .. x2 = col_double(),
## .. x3 = col_double(),
## .. x4 = col_double(),
## .. x5 = col_double(),
## .. x6 = col_double(),
## .. x7 = col_double(),
## .. x8 = col_double(),
## .. x9 = col_double()

```

```
##    .. )
##    - attr(*, "problems")=<externalptr>

#measures of central tendencies can only be computed for the numerical variables. i

#describe(Depr)
knitr::kable(describe(Depr), booktabs=TRUE, longtable=T, caption = "Descriptive sta
kableExtra::kable_styling(full_width = FALSE)
```

[t]rrrrrrrrrrr Descriptive statistics

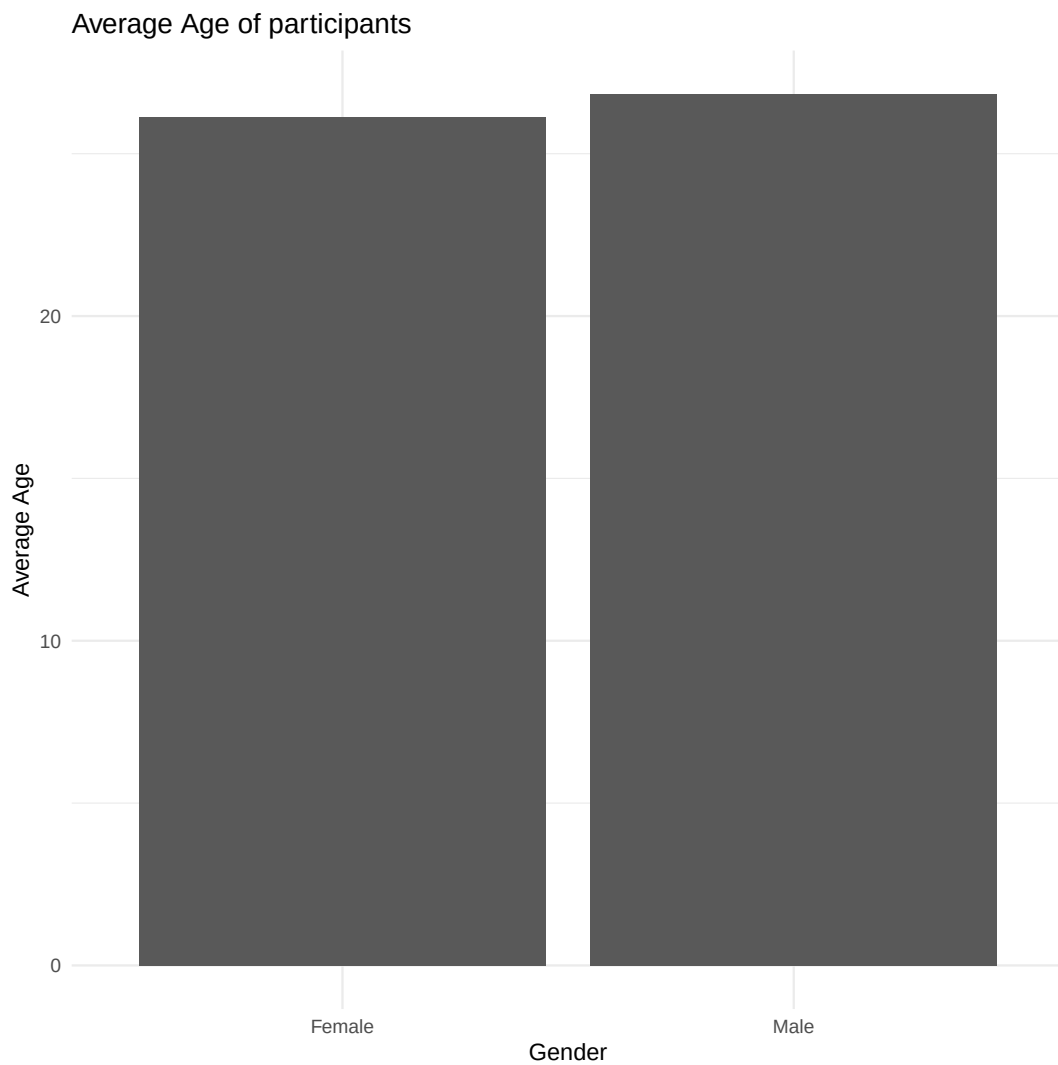
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Age	1	247	26.348178	6.7823470	24	26.1457286	5.9304	12	40	28	0.3067367	-0.8428786	0.4315504
x1	2	247	1.704453	1.0229582	1	1.7537688	1.4826	0	3	3	0.0439465	-1.3143077	0.0650893
x2	3	247	1.736842	1.0551745	2	1.7939698	1.4826	0	3	3	-0.1043000	-1.3142945	0.0671391
x3	4	247	1.732794	1.1340344	2	1.7889447	1.4826	0	3	3	-0.1823403	-1.4259668	0.0721569
x4	5	247	2.016194	1.0358124	2	2.1306533	1.4826	0	3	3	-0.5130218	-1.1123751	0.0659072
x5	6	247	1.336032	1.1245764	1	1.2964824	1.4826	0	3	3	0.2732781	-1.3089854	0.0715551
x6	7	247	1.615385	1.1270177	1	1.6432161	1.4826	0	3	3	-0.0068279	-1.4277033	0.0717104
x7	8	247	1.368421	1.0959528	1	1.3366834	1.4826	0	3	3	0.2339658	-1.2609069	0.0697338
x8	9	247	1.093117	0.9770919	1	0.9949749	1.4826	0	3	3	0.5173034	-0.7596885	0.0621709
x9	10	247	0.951417	1.0958477	1	0.8190955	1.4826	0	3	3	0.8339846	-0.6853221	0.0697271

- data Visualization

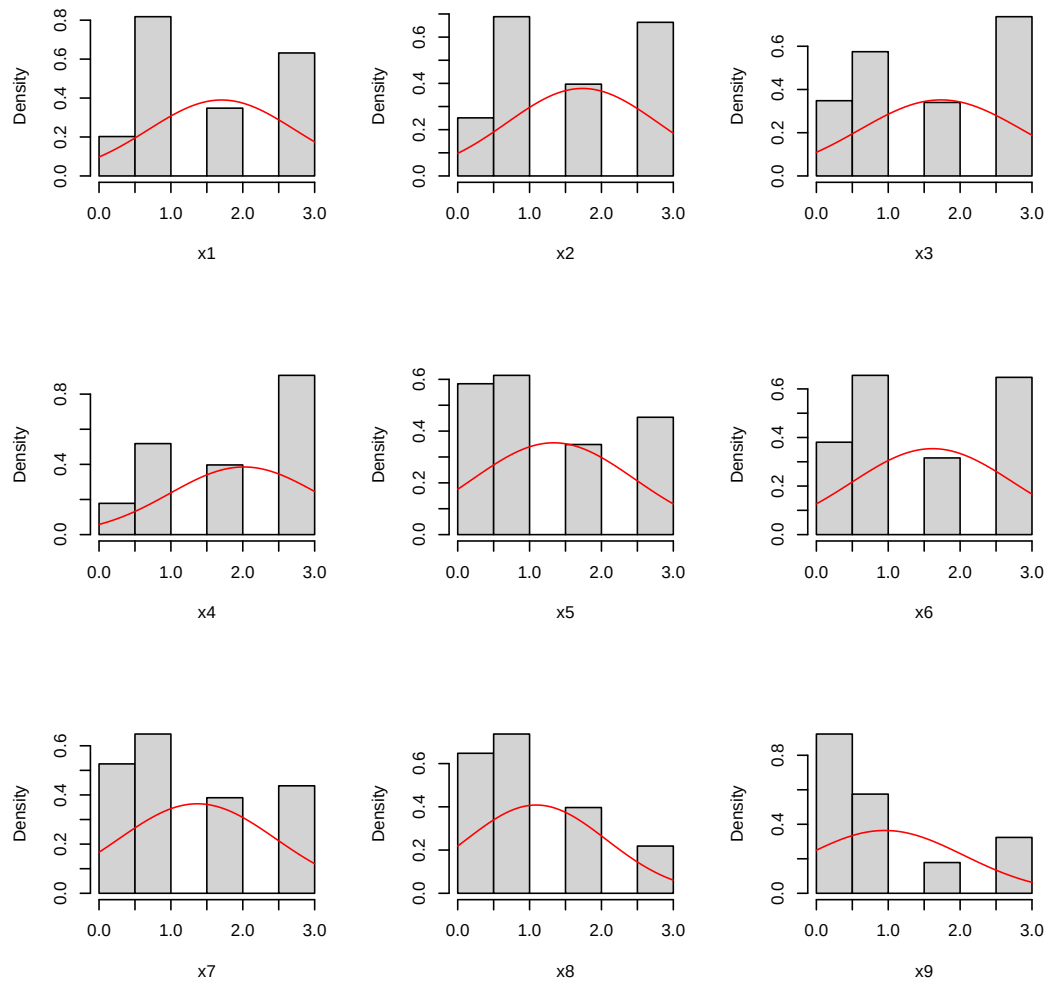
```
Depression$Gender[Depression$Gender==1]<-"Male"
Depression$Gender[Depression$Gender==2]<-"Female"
View(Depression)

#barplot of count by gender
by_gender<-Depression %>%
  group_by(Gender) %>%
  summarize(mean_age=mean(Age))

ggplot(by_gender, aes(x=Gender, y=mean_age), fill=Gender) +
  geom_col() +
  labs(x="Gender", y="Average Age", title = "Average Age of participants") +
  theme_minimal()
```



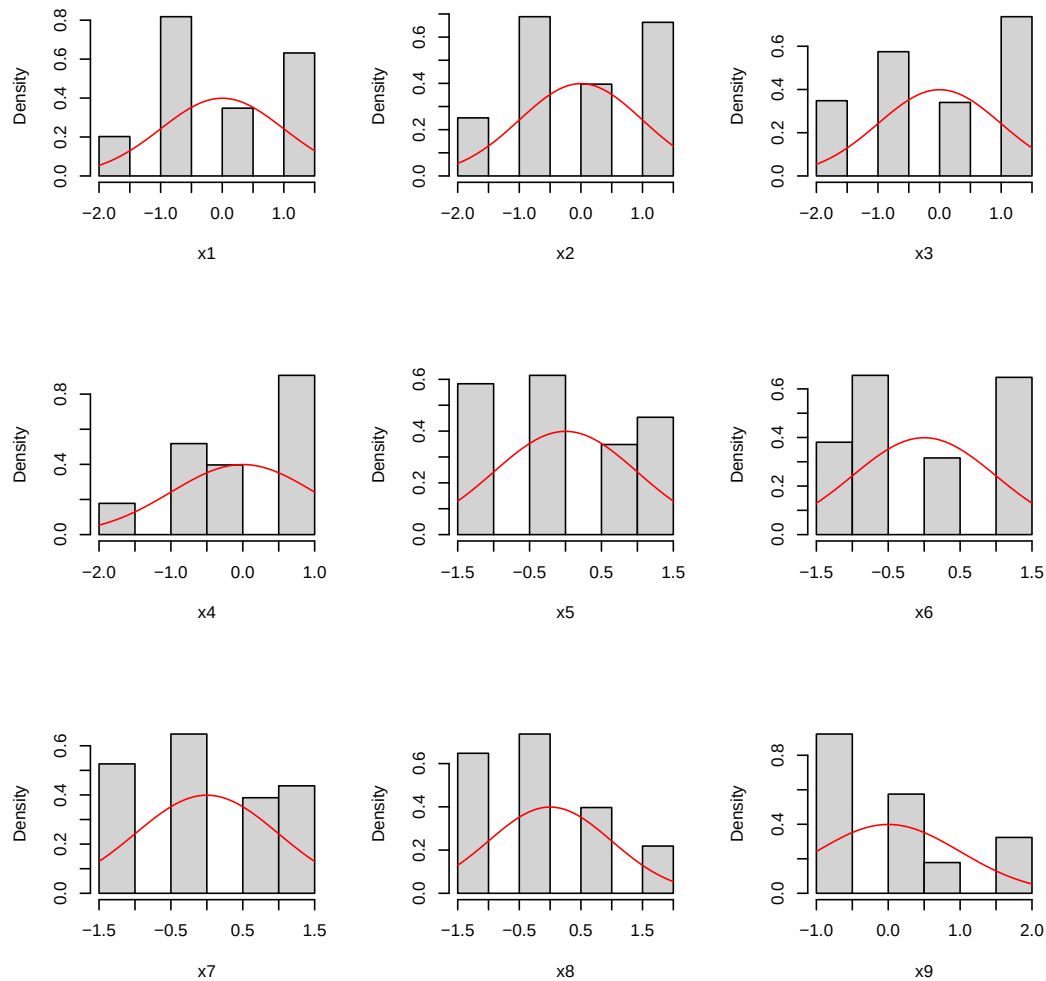
```
#ggplot (Depression, aes (x=Gender, y=freq), fill=Gender) +  
#   geom_bar("identity") +  
#   labs(x="Gender", y="count", title = "count of males and females") +  
#   theme_minimal()  
  
library (MVN)  
ind_var <- df[,paste("x", 1:9, sep="")]  
normality1<-mvn(data=ind_var, mvnTest="royston", univariatePlot="histogram")
```



variables x2,x4, x8,x9 are not normally distributed. However we have no serious deviations. To achieve normality, we will standardize our data when conducting Principal components analysis.

- data Normalization

```
Zscore<-scale(data_dep)
View(Zscore)
normality2<-mvn(data=Zscore, mvnTest="royston", univariatePlot="histogram")
```



```
normality2
```

```
## $multivariateNormality
##      Test      H      p value MVN
## 1 Royston 594.5605 5.030746e-122 NO
##
## $univariateNormality
##      Test Variable Statistic  p value Normality
## 1 Anderson-Darling x1      17.5696 <0.001 NO
## 2 Anderson-Darling x2      15.1837 <0.001 NO
## 3 Anderson-Darling x3      15.9452 <0.001 NO
## 4 Anderson-Darling x4      19.7831 <0.001 NO
## 5 Anderson-Darling x5      13.7992 <0.001 NO
## 6 Anderson-Darling x6      15.1708 <0.001 NO
## 7 Anderson-Darling x7      12.8536 <0.001 NO
## 8 Anderson-Darling x8      14.0168 <0.001 NO
## 9 Anderson-Darling x9      22.1000 <0.001 NO
##
## $Descriptives
##      n      Mean Std.Dev      Median      Min      Max      25th
## x1 247 -3.731060e-19      1 -0.68864342 -1.6662005 1.2664707 -0.6886434
## x2 247 -9.116077e-18      1  0.24939751 -1.6460236 1.1971080 -0.6983130
## x3 247  4.613083e-17      1  0.23562467 -1.5279903 1.1174321 -0.6461828
## x4 247 -1.386534e-16      1 -0.01563443 -1.9464859 0.9497913 -0.9810602
## x5 247  1.803638e-18      1 -0.29880798 -1.1880317 1.4796395 -1.1880317
## x6 247 -3.208229e-17      1 -0.54602925 -1.4333268 1.2285658 -0.5460292
## x7 247 -7.647938e-17      1 -0.33616507 -1.2486131 1.4887310 -1.2486131
## x8 247 -8.811623e-17      1 -0.09530057 -1.1187458 1.9515899 -1.1187458
```

```
## x9 247 2.173672e-17 1 0.04433371 -0.8682019 1.8694049 -0.8682019
## 75th Skew Kurtosis
## x1 1.2664707 0.043946482 -1.3143077
## x2 1.1971080 -0.104299967 -1.3142945
## x3 1.1174321 -0.182340330 -1.4259668
## x4 0.9497913 -0.513021751 -1.1123751
## x5 0.5904158 0.273278054 -1.3089854
## x6 1.2285658 -0.006827855 -1.4277033
## x7 0.5762830 0.233965842 -1.2609069
## x8 0.9281446 0.517303404 -0.7596885
## x9 0.5006015 0.833984627 -0.6853221
```

itemize)

5 Exploratory factor analysis

```
#Exploratory factor analysis
bart_spher(data_dep)

## Bartlett's Test of Sphericity
##
## Call: bart_spher(x = data_dep)
##
## X2 = 844.617
## df = 36
## p-value < 2.22e-16
```

produces Bartlett's test of sphericity to understand if the model will be significant

p-value is less than 0.05 hence the test is significant

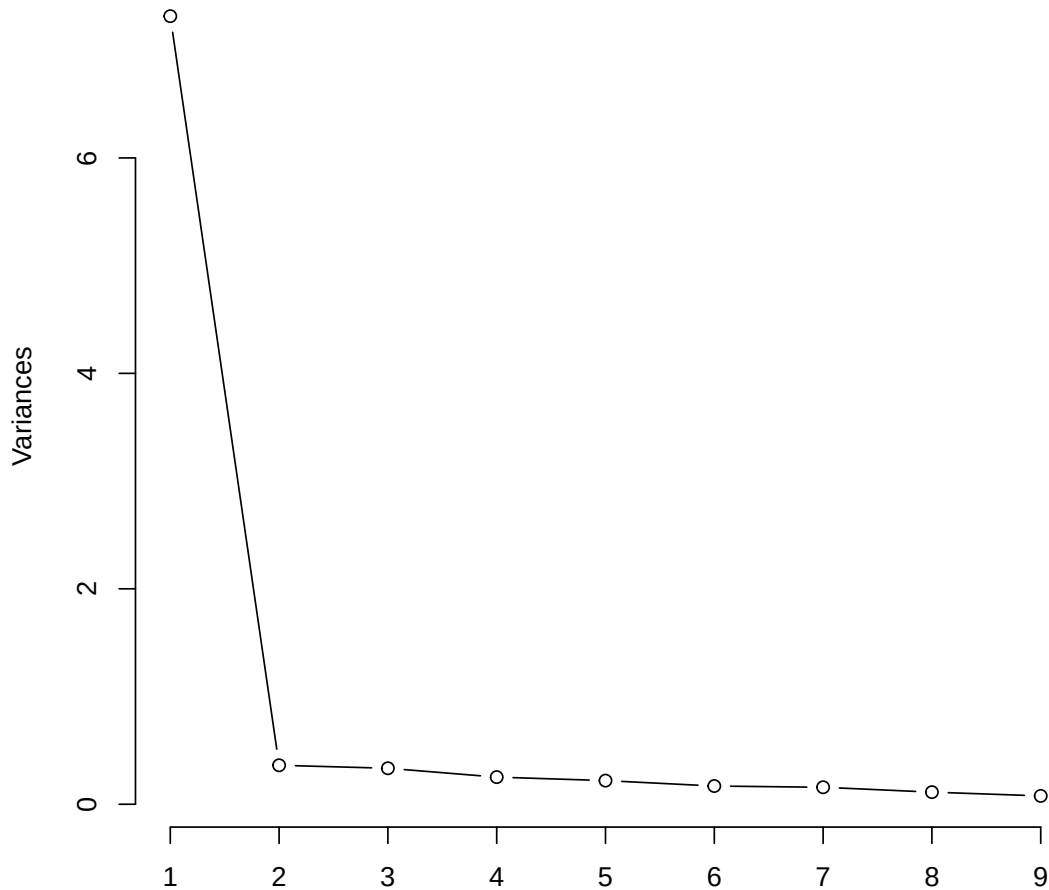
6 Principal components analysis

```
#Performing PCA using prcomp() function
pca<-prcomp(data_dep, retx=TRUE, center=FALSE, scale=TRUE)
summary(pca)

## Importance of components:
##
## PC1 PC2 PC3 PC4 PC5 PC6 PC7
## Standard deviation 2.7049 0.60143 0.57790 0.50191 0.46854 0.41118 0.39716
## Proportion of Variance 0.8129 0.04019 0.03711 0.02799 0.02439 0.01879 0.01753
## Cumulative Proportion 0.8129 0.85310 0.89021 0.91820 0.94259 0.96138 0.97891
##
## PC8 PC9
## Standard deviation 0.33520 0.27836
## Proportion of Variance 0.01248 0.00861
## Cumulative Proportion 0.99139 1.00000

screeplot(pca, type='l')
```

pca



```
sd <-pca$sdev

loadings <- pca$rotation

#rownames(loadings) <- colnames(R)
scores <- pca$x
```

6.1 Interpretation of PCA output

Principal Component Analysis (PCA) is a dimensionality reduction technique commonly used in data analysis and visualization.

- Importance of Components:

The summary typically starts with a table that shows the proportion of variance explained by each principal component (PC) and the cumulative proportion of variance explained. It indicates the importance of each PC in capturing the variability in the data. Higher values suggest that the corresponding PC accounts for more information in the dataset.

```
summary(pca)

## Importance of components:
##
## Standard deviation      PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Proportion of Variance 0.8129 0.04019 0.03711 0.02799 0.02439 0.01879 0.01753
## Cumulative Proportion 0.8129 0.85310 0.89021 0.91820 0.94259 0.96138 0.97891
```



```
##              PC8      PC9
## Standard deviation 0.33520 0.27836
## Proportion of Variance 0.01248 0.00861
## Cumulative Proportion 0.99139 1.00000
```

- **Standard Deviations:**

The next table displays the standard deviation of each principal component. It represents the spread of scores along each PC. Larger standard deviations suggest that the PC captures more variability in the original data.

```
knitr::kable(pca$sdev, booktabs=TRUE, longtable=T, caption = "Standard Deviations",
  kableExtra::kable_styling(full_width = FALSE))
```

[t]r Standard Deviations

x
2.7048513
0.6014283
0.5779012
0.5019126
0.4685424
0.4111791
0.3971576
0.3351967
0.2783635

- **Proportion of Variance:**

This table provides the proportion of variance explained by each PC. It helps assess the contribution of individual PCs to the overall variance in the dataset. Higher proportions indicate that the corresponding PC retains a significant amount of information from the original data.

```
summary(pca)

## Importance of components:
##              PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation 2.7049 0.60143 0.57790 0.50191 0.46854 0.41118 0.39716
## Proportion of Variance 0.8129 0.04019 0.03711 0.02799 0.02439 0.01879 0.01753
## Cumulative Proportion 0.8129 0.85310 0.89021 0.91820 0.94259 0.96138 0.97891
##              PC8      PC9
## Standard deviation 0.33520 0.27836
## Proportion of Variance 0.01248 0.00861
## Cumulative Proportion 0.99139 1.00000
```

- **Cumulative Proportion of Variance:**

Similar to the importance of components table, this table presents the cumulative proportion of variance explained by the PCs. It allows you to understand how much total variance is accounted for a given number of PCs. Typically, you aim to retain a cumulative proportion close to 1 (70% is typically chosen) to capture a substantial amount of variability.

```
summary(pca)

## Importance of components:
##              PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation 2.7049 0.60143 0.57790 0.50191 0.46854 0.41118 0.39716
## Proportion of Variance 0.8129 0.04019 0.03711 0.02799 0.02439 0.01879 0.01753
## Cumulative Proportion 0.8129 0.85310 0.89021 0.91820 0.94259 0.96138 0.97891
##              PC8      PC9
## Standard deviation 0.33520 0.27836
## Proportion of Variance 0.01248 0.00861
```

```
## Cumulative Proportion 0.99139 1.00000
```

- Loadings:

The loadings section shows the correlation between the original variables and each principal component. It represents the contribution of each variable to each PC. Positive or negative values indicate the direction and strength of the relationship. Variables with higher absolute loading values have a more significant impact on the corresponding PC.

```
#pca$rotation
knitr::kable(pca$rotation, booktabs=TRUE, longtable=T, caption = "loadings",) %>%
  kableExtra::kable_styling(full_width = FALSE)
```

[t]rrrrrrrr loadings

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
x1	-0.3313213	0.2405843	-0.1518165	0.6489980	-0.3192214	0.2295026	0.3206828	0.3595045	0.0380008
x2	-0.3507561	-0.0358480	-0.1394410	0.1908451	-0.0046149	0.1136319	-0.1482687	-0.7039229	0.5379544
x3	-0.3373868	0.2173807	-0.3078203	-0.0497565	0.0880032	-0.7466463	-0.2636625	0.2746371	0.1775330
x4	-0.3525266	0.1940792	-0.1421475	0.0348539	0.0429909	-0.0692589	0.0084622	-0.4186717	-0.7966398
x5	-0.3285995	0.1422229	0.0905325	-0.6147897	-0.6769830	0.1452959	0.0024065	0.0473857	0.0629644
x6	-0.3413658	-0.0916276	-0.0927938	-0.0710164	0.3266536	0.5276328	-0.6011698	0.3322899	-0.0670755
x7	-0.3369660	0.0080996	-0.1093994	-0.3594906	0.5221837	0.1136792	0.6588999	0.0752952	0.1406034
x8	-0.3106558	0.1558649	0.9012510	0.1341881	0.1544741	-0.1477488	-0.0436953	0.0146602	0.0335123
x9	-0.3074577	-0.8960639	0.0389900	0.0918673	-0.1641254	-0.1946249	0.0901117	0.0774030	-0.1168421

- Scores:

The scores table displays the transformed scores for each observation (data point) along each PC. These scores represent the coordinates of each observation in the reduced-dimensional space spanned by the PCs. They can be used for data visualization or further analysis.

```
Scores<-pca$x
View(Scores)
knitr::kable(head(Scores,5), booktabs=TRUE, longtable=T, caption = "Scores",) %>%
  kableExtra::kable_styling(full_width = FALSE)
```

[t]rrrrrrrr Scores

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
	-0.5333434	0.3125191	0.4751417	0.4327655	-0.0361688	-0.0158854	0.1350380	0.0061590	-0.3089697
	-2.1348959	0.8173846	0.7291599	0.4706794	-0.6546239	0.5110440	-0.3793972	0.5550213	0.1059709
	-2.1452953	0.0194213	0.2836785	-0.5457575	-0.9204338	0.2997469	-0.6046450	0.0798565	-0.3503551
	-1.5795556	0.2649735	0.0547354	0.1897301	0.7983219	0.5472128	-0.6039724	0.3410356	0.0190303
	-3.4536945	-0.9934925	-0.1646720	0.0070043	-0.9592909	-0.0086248	-0.4224186	1.0467741	0.7103359

itemize)

7 Performing Factor Analysis using factanal () function.

```
#Performing Factor Analysis FA using factanal() function
#Maximum Likelihood (MLE) approach
R<-cov(data_dep)
factanal(covmat=R,factors=1,method="mle",n.obs=247,rotation = "none")

##
## Call:
## factanal(factors = 1, covmat = R, n.obs = 247, rotation = "none", method = "mle")
##
## Uniquenesses:
##      x1      x2      x3      x4      x5      x6      x7      x8      x9
## 0.748 0.377 0.593 0.391 0.590 0.500 0.470 0.776 0.575
##
## Loadings:
##      Factor1
## x1 0.502
## x2 0.789
## x3 0.638
## x4 0.780
## x5 0.640
## x6 0.707
## x7 0.728
## x8 0.473
## x9 0.652
##
##
##              Factor1
## SS loadings      3.980
## Proportion Var   0.442
##
## Test of the hypothesis that 1 factor is sufficient.
## The chi square statistic is 44.51 on 27 degrees of freedom.
## The p-value is 0.0183

factanal(covmat=R,factors=2,method="mle",n.obs=247,rotation = "none")

##
## Call:
## factanal(factors = 2, covmat = R, n.obs = 247, rotation = "none", method = "mle")
##
## Uniquenesses:
##      x1      x2      x3      x4      x5      x6      x7      x8      x9
## 0.750 0.366 0.564 0.261 0.596 0.478 0.477 0.773 0.429
##
## Loadings:
##      Factor1 Factor2
## x1  0.500
## x2  0.788  0.111
## x3  0.641 -0.160
## x4  0.813 -0.280
## x5  0.635
## x6  0.701  0.178
## x7  0.723
## x8  0.467
## x9  0.661  0.365
##
##
##              Factor1 Factor2
## SS loadings      4.015  0.292
## Proportion Var   0.446  0.032
## Cumulative Var   0.446  0.478
##
## Test of the hypothesis that 2 factors are sufficient.
```

```
## The chi square statistic is 21.96 on 19 degrees of freedom.
## The p-value is 0.286

factanal(covmat=R, factors=5, method="mle", n.obs=247, rotation = "none")

##
## Call:
## factanal(factors = 5, covmat = R, n.obs = 247, rotation = "none", method = "mle")
##
## Uniquenesses:
##      x1      x2      x3      x4      x5      x6      x7      x8      x9
## 0.690 0.233 0.411 0.005 0.548 0.398 0.434 0.578 0.511
##
## Loadings:
##      Factor1 Factor2 Factor3 Factor4 Factor5
## x1  0.418    0.278  -0.144          0.193
## x2  0.638    0.538  -0.262          0.113
## x3  0.569    0.286   0.195  -0.364    0.113
## x4  0.997
## x5  0.527    0.334   0.198          0.138
## x6  0.519    0.520   0.104        -0.222
## x7  0.603    0.379   0.227
## x8  0.360    0.322   0.184   0.339   0.201
## x9  0.446    0.531
##
##
##              Factor1 Factor2 Factor3 Factor4 Factor5
## SS loadings      3.140   1.360   0.263   0.260   0.168
## Proportion Var    0.349   0.151   0.029   0.029   0.019
## Cumulative Var    0.349   0.500   0.529   0.558   0.577
##
## Test of the hypothesis that 5 factors are sufficient.
## The chi square statistic is 0 on 1 degree of freedom.
## The p-value is 0.957

#note that two factor model is significant to a maximum of 5 factors
```

8 Performing Factor Analysis using principal components

```
data_cov<-cov(data_dep)
fa <- principal(data_dep, nfactors=2, rotate="varimax")
print(fa, digits=2, cutoff=.3, sort=TRUE)

## Principal Components Analysis
## Call: principal(r = data_dep, nfactors = 2, rotate = "varimax")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      item RC1  RC2  h2  u2 com
## x3      3 0.76 0.10 0.59 0.41 1.0
## x4      4 0.76 0.32 0.67 0.33 1.3
## x2      2 0.73 0.36 0.67 0.33 1.5
## x1      1 0.63 0.07 0.41 0.59 1.0
## x7      7 0.62 0.44 0.58 0.42 1.8
## x6      6 0.58 0.48 0.56 0.44 1.9
## x9      9 0.51 0.49 0.51 0.49 2.0
## x8      8 0.04 0.90 0.82 0.18 1.0
## x5      5 0.49 0.52 0.51 0.49 2.0
##
##
##              RC1  RC2
## SS loadings    3.31 2.01
## Proportion Var  0.37 0.22
## Cumulative Var  0.37 0.59
```

```

## Proportion Explained  0.62 0.38
## Cumulative Proportion 0.62 1.00
##
## Mean item complexity =  1.5
## Test of the hypothesis that 2 components are sufficient.
##
## The root mean square of the residuals (RMSR) is  0.08
## with the empirical chi square 118.03 with prob < 2.6e-16
## 0.3
## Fit based upon off diagonal values = 0.97

fa$loadings

##
## Loadings:
##      RC1   RC2
## x1 0.635
## x2 0.734 0.364
## x3 0.761 0.101
## x4 0.755 0.319
## x5 0.486 0.525
## x6 0.577 0.478
## x7 0.623 0.443
## x8           0.902
## x9 0.513 0.494
##
##              RC1   RC2
## SS loadings    3.314 2.007
## Proportion Var 0.368 0.223
## Cumulative Var 0.368 0.591

fa$fit

## [1] 0.9090903

fa$rotation

## [1] "varimax"

fa$communality

##      x1      x2      x3      x4      x5      x6      x7      x8
## 0.4074071 0.6717778 0.5890017 0.6721469 0.5115774 0.5612489 0.5845545 0.8150551
##      x9
## 0.5078730

fa$STATISTIC

## [1] 111.6503

fa$fn

## [1] "principal"

fa$Call

## principal(r = data_dep, nfactors = 2, rotate = "varimax")

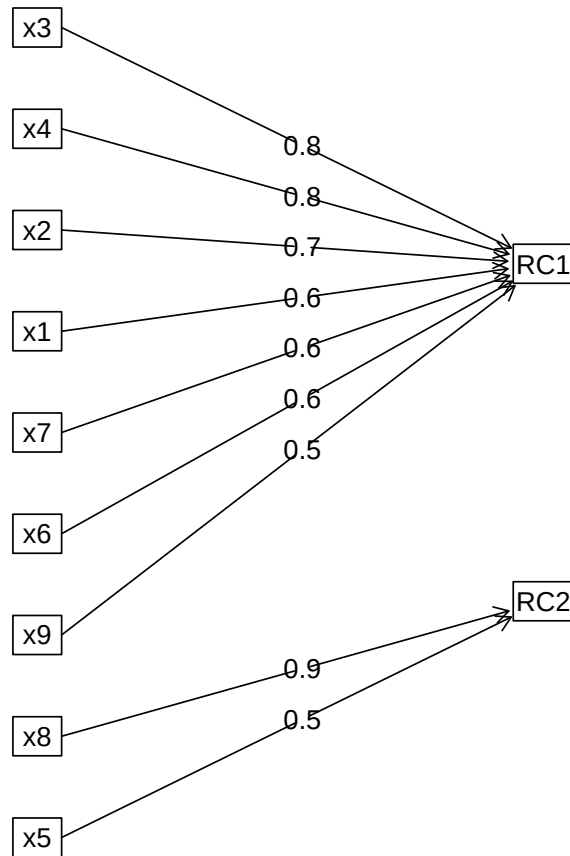
fa$uniquenesses

##      x1      x2      x3      x4      x5      x6      x7      x8
## 0.5925929 0.3282222 0.4109983 0.3278531 0.4884226 0.4387511 0.4154455 0.1849449
##      x9
## 0.4921270

fa.diagram(fa)

```

Components Analysis



8.1 Interpretation of FA model output

- **Model Fit:** The output often includes information about the overall model fit. It may include statistics such as the chi-square statistic, degrees of freedom, and p-value. These values assess the adequacy of the model fit to the data. A non-significant p-value suggests a good fit, indicating that the model adequately explains the observed data.

```
summary(fa)
##
## Factor analysis with Call: principal(r = data_dep, nfactors = 2, rotate = "varimax")
##
## Test of the hypothesis that 2 factors are sufficient.
## The degrees of freedom for the model is 19 and the objective function was 0.46
## The number of observations was 247 with Chi Square = 111.65 with prob < 4e-16
##
## The root mean square of the residuals (RMSA) is 0.08
```

- **Factor Loadings:**

Factor loadings represent the relationships between observed variables and underlying latent factors. The output provides a table or matrix displaying the loadings for each variable on each factor. Higher absolute values indicate stronger associations. Interpretation of loadings involves identifying variables that have significant loadings on specific factors, which helps understand the latent constructs captured by those factors.

```
fa$loadings
##
```

```
## Loadings:
##      RC1    RC2
## x1 0.635
## x2 0.734 0.364
## x3 0.761 0.101
## x4 0.755 0.319
## x5 0.486 0.525
## x6 0.577 0.478
## x7 0.623 0.443
## x8      0.902
## x9 0.513 0.494
##
##              RC1    RC2
## SS loadings   3.314 2.007
## Proportion Var 0.368 0.223
## Cumulative Var 0.368 0.591
```

- **Eigenvalues:**

Eigenvalues represent the amount of variance explained by each factor. The output typically provides a table showing the eigenvalues and the proportion of variance explained by each factor. Higher eigenvalues indicate factors that capture more variability in the data. Typically eigenvalues greater than 1 or those that contribute a significant proportion of the variance are chosen.

```
varcov<-var(data_dep)
eigen(varcov)$values
## [1] 5.2600462 0.9004416 0.8484151 0.8132411 0.6326914 0.6029885 0.5328581
## [8] 0.4755238 0.3441164
```

- **Proportion of Variance Explained:**

This component shows the proportion of total variance explained by each factor. It helps assess the relative contribution of each factor to the overall variability in the dataset. Higher proportions suggest that the corresponding factor explains a larger portion of the variance.

```
fa$Vaccounted
##              RC1    RC2
## SS loadings   3.3136480 2.0069943
## Proportion Var 0.3681831 0.2229994
## Cumulative Var 0.3681831 0.5911825
## Proportion Explained 0.6227910 0.3772090
## Cumulative Proportion 0.6227910 1.0000000
knitr::kable(fa$Vaccounted, booktabs=TRUE, longtable=T, caption = "Proportion of Variance Explained",
  kableExtra::kable_styling(full_width = FALSE))
```

[t]lrr Proportion of Variance Explained

	RC1	RC2
--	-----	-----

SS loadings	3.3136480	2.0069943
Proportion Var	0.3681831	0.2229994
Cumulative Var	0.3681831	0.5911825
Proportion Explained	0.6227910	0.3772090
Cumulative Proportion	0.6227910	1.0000000

@

- **Scree Plot:**

A scree plot is a graphical representation of the eigenvalues. It helps determine the number of factors to retain in the analysis. The scree plot typically displays the eigenvalues on the y-axis and the factor number on the x-axis. The plot shows a steep drop-off followed by a more gradual decline. The number of factors to retain is usually determined by examining the point where the drop-off levels off or where the slope becomes less significant.

#refer to the scree plot at the PCA analysis

- **Factor Correlations:**

The output often includes a correlation matrix that shows the relationships between the extracted factors.

Positive correlations suggest that the factors are related or capture similar constructs, while negative correlations indicate that the factors are orthogonal or capture distinct constructs.

```
fa_cor<-fa$scores
View(fa_cor)
knitr::kable(head(fa_cor,10), booktabs=TRUE, longtable=T,caption = "Factor correlations",
  kableExtra::kable_styling(full_width = FALSE))
```

[t]rr Factor correlations

RC1 RC2	
-1.7351538	-0.2505527
-0.7943403	0.5721376
-0.6820944	0.5604937
-0.8312209	0.0258253
0.5801439	0.3599470
-0.9558495	-1.1047408
-0.5188204	0.2470050
-0.0721557	-0.2370873
-0.2922981	-1.2971812
-0.9127804	-0.0334856

Interpreting the factor model output in R involves considering these various components to understand the relationships between observed variables and latent factors, assess the adequacy of the model fit, determine the number of factors to retain, and gain insights into the underlying structure of the data.